
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

1 Language models are often confidently wrong. We ask whether a model’s
2 activations carry a better signal of its own errors than its stated confidence
3 does. We test four open models on four tasks: MMLU-Pro, MATH, GPQA-
4 Diamond, and a geometry task graded by a compiler. We put a linear probe
5 on the hidden state to predict whether an answer is right. This probe was
6 better than the model’s stated confidence in 12 of 16 cases, by 0.09 AUROC
7 on average. The gap is largest on MATH. The probe is not just detecting
8 hard questions: it still works when the same question is attempted five times.
9 The signal exists before the model is asked how confident it is. On Mistral,
10 the model uses this signal when it reports confidence. When we erase the
11 signal from the hidden state, the stated confidence no longer separates right
12 answers from wrong ($0.84 \rightarrow 0.56$), while erasing a random direction did
13 nothing. On GLM and Qwen3.6, erasing it changes nothing. In short, four
14 open models carry more confidence signal about their own errors than they
15 report. One of them uses that information when it self-reports.

16 1 Introduction

17 A language model that is confidently wrong is a problem. Its stated confidence, which we
18 call its self-report, is a signal for deciding whether to trust its answer or not. This paper
19 asks whether the model’s activations contain a signal related to its confidence.

20 We have two hypotheses. Upon giving an incorrect answer and saying it has high confidence,
21 either the model genuinely has no idea it might be wrong and is just blindly confident, or
22 holds the error signal deep down in its activations but does not surface it in its self-report.
23 Hence it is either a capability problem or a reporting problem.

24 In this paper we tell these two apart. We read the model’s hidden state with a linear *probe*
25 and compare what the probe finds to what the model reports. We say a fact is *represented*
26 when a probe can read that fact from the model’s state. We say a fact is *reported* when that
27 fact reaches the confidence number the model writes. We refer to *knowing* as shorthand for
28 a representation used by self-report. We make no claim about awareness. A probe alone
29 could not support one.

30 Prior work reads the truth of a statement from activations [1, 2, 9], and reads a model’s own
31 errors from its hidden state better than its output reveals them [3, 10]. A model’s output
32 probabilities are often better calibrated than its words [5], and asked confidence is known
33 to be overconfident [8, 11, 12]. Probes can latch onto confounds rather than the intended
34 variable [6]. Intervening on a truth direction changes what a model says [7, 13]. We apply
35 those tools to the model’s own attempts, and then ask whether the self-report depends on
36 the representation.

2 Reading the state

Models and tasks. We test four open models of three architectures: Mistral-Small-24B (dense), Qwen3.6-27B (gated linear attention interleaved with full attention), and GLM-4.7-Flash and Gemma-4-26B (mixture of experts). Each answers 150 questions, five times each, in four domains: MMLU-Pro, MATH, GPQA-Diamond, and GeoGenBench. GeoGenBench is a geometry task that uses a compiler to check the constructions, so its labels need no judge. Four models times four tasks gives 16 cells, of about 750 attempts each.

Three-step prompting structure.

1. The model is shown the question and asked, before answering, how confident it is, from 0 to 100, that it will get it right.
2. It answers.
3. It is asked how confident it is that the answer it just gave is correct.

The probe. At the moment the model is about to write its confidence digit, we record the number and the residual-stream hidden state at that token (2,048 to 5,120 dimensions, depending on the model). The probe is a linear classifier on that vector: features are standardized, reduced to 50 principal components, and fed to an L2-regularized logistic regression that predicts whether the attempt was correct. We fit it with five-fold cross-validation grouped by question, so all five attempts at a question fall in the same fold and the probe is always scored on questions it never saw. The read depth is fixed in advance at 70% of the network (Appendix E). Every readout is scored with AUROC.

The activation state carries more than self-report. The probe beats the model’s self-report in **12 of 16** cells, with one tie. The mean gain is +0.09 AUROC (Appendix D). The gap is largest on MATH, at +0.20. Mistral on MATH is a good example of what this looks like. The model is never told whether it got the answer right. Yet after attempting, it raises its confidence, and it raises it more on the attempts it got wrong (+22 points) than on the ones it got right (+16 points). The probe on the same records reads 0.83. The state tracks correctness better than the self-report does.

Why MATH. On MATH, right and wrong answers have the same shape. Both are a chain of steps ending in a boxed result, so the model cannot tell from the look of its answer whether it went wrong. Appendix A follows one attempt where a single wrong subtraction leads to a wrong answer, and the model then raises its confidence. Shape still carries a little signal. A baseline classifier that sees only surface features of the answer text, such as its length and line count, reaches 0.74, but the probe reaches 0.83. Geometry is the opposite case. A failed construction usually does not compile, so the failure is visible in the answer itself, and there the surface baseline does as well as the probe. The internal signal helps most where the failure is invisible from outside.

Controls. Three things could make a probe look better than it is; the next section tests each. It could be reading how hard the question is rather than the attempt, so we compare attempts within one question. It could be reading the shape of the output, so we compare it with the surface baseline in every cell; on Mistral, adding the probe to the surface features raises AUROC by +0.21 on MMLU-Pro, +0.07 on MATH, +0.05 on GPQA, and −0.11 on geometry. And it could be reading something that was there before the attempt, so we check the input layer, where the token is identical in every record and the probe sits at exactly 0.5. These controls were run on Mistral only, for time. Its four cells reproduce the headline.

3 What the signal is

Difficulty control. A probe that beats self-report could be doing something simpler. It could be learning which questions are hard. Hard questions fail more often, so a difficulty detector would predict correctness without reading the attempt. We run two checks.

Table 1: Mistral. Rows: training domain; columns: test domain; bold: in-domain.

train ↓ test →	GeoGen	MMLU-Pro	MATH	GPQA
GeoGen	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

Table 2: Mistral on MATH, 150 held-out records. The correctness direction scaled by a gain at the confidence token, after the answer is written. Bold marks the erased and quadrupled settings; unparseable means the output no longer parsed as a confidence.

	untouched	erased	doubled	quadrupled
gain on the correctness direction	1	0	2	4
mean confidence, wrong / right answers	84 / 96	97 / 98	70 / 94	54 / 88
AUROC of self-report	0.84	0.56	0.85	0.82
same, with a random direction	0.84	0.83	0.84	unparseable

85 First, hold the question fixed. Each question is attempted five times, so within one question,
 86 difficulty cannot vary. The probe still separates the successful attempts from the failed ones
 87 (0.72 to 0.74 in the cells of Table 3 in Appendix D). The model’s own $P(\text{True})$, its probability
 88 of answering “True” when asked whether it was right, drops to near chance (0.47 to 0.59).

89 Second, ask when the direction appears. Before the attempt, all five tries at a question look
 90 identical to the model. Anything readable at that moment is about the question, not the
 91 attempt. A difficulty detector would still work there. Our probe, trained on the state after
 92 the attempt, reads 0.49 at that pre-attempt point on MMLU-Pro, which is chance, against
 93 0.75 at the post-attempt token where it was trained. Geometry is the exception. There the
 94 pre-attempt state already predicts failure, so that cell is mostly difficulty.

95 **Reading before the question.** A subtler worry. Every read so far came after the model
 96 was asked to assess itself. Maybe the question creates the signal rather than reveals it. So
 97 we read the state at the last token of the answer, before any confidence question exists.
 98 This test can only work where the question alone predicts nothing. That rules out MATH,
 99 where the question alone already predicts correctness at 0.79, and GPQA, where nothing is
 100 readable at any site. On MMLU-Pro the question alone reads 0.57, chance. At the end of
 101 the answer, correctness reads 0.69. Asking the model to self-assess sharpens it to 0.77.

102 **Transfer across domains.** Table 1 trains a probe on one domain and tests it on each
 103 of the others. On Mistral it keeps about 90% of its lift above chance (0.69 off the diagonal
 104 against 0.71 on it). The full four-model matrix showed the same pattern. The one exception
 105 is Gemma-4, where probes trained on geometry do not transfer out, though probes trained
 106 elsewhere transfer in.

107 4 Whether self-report depends on the residual signal

108 **Erasing and amplifying the signal direction.** A model could carry a perfect internal
 109 record of its errors and never consult it during self-report. The only way to find out is to
 110 change the record and watch the number. At the confidence token, after the answer is written
 111 so the answer cannot change, we scale the model’s position along the correctness direction
 112 by a gain g . Gain 1 leaves the model untouched, gain 0 erases the per-attempt information,
 113 and gains 2 and 4 exaggerate it. We run this on Mistral and MATH, 150 held-out records,
 114 with a random direction of the same size as a control. Table 2 shows what the model wrote
 115 at each setting.

116 **Result on Mistral.** Self-report loses its information when the signal direction is erased.
 117 Its AUROC falls from 0.84 to 0.56, a drop of 0.27 (95% interval 0.12 to 0.42). Erase a

random direction and nothing happens. This is not generic damage. The output stays well formed. Confidence on wrong answers goes *up* to match right answers, so the model becomes uniformly sure rather than confused. Amplify the direction and the opposite happens: confidence on wrong answers falls while right answers hold, and the expected calibration error halves, $0.34 \rightarrow 0.19$. Two checks say the effect is specific. The same intervention at 30% of the way through the network does nothing. A direction learned on MMLU-Pro, where surface features cannot explain the probe, still controls the MATH report ($0.88 \rightarrow 0.71$).

Results on GLM and Qwen3.6. GLM has the strongest signal in the study: within a question on MATH, the probe reads 0.87 while the model’s own $P(\text{True})$ reads 0.53. Yet erasing the direction leaves self-report where it was, 0.76 before and after. Qwen3.6 is a third case. Its self-report on MATH is nearly uninformative to begin with (AUROC 0.54; it writes 97 or 98 on right and wrong answers alike), so erasing the direction has nothing to remove. Amplifying the direction fourfold raises the self-report’s AUROC to 0.63, while a random direction leaves it at 0.54. Both models represent their errors and report them poorly. These are our cleanest cases of represented but not reported, and they make the causal result a fact about particular models, not models in general.

5 The Jacobian lens

A label-free check. A trained probe might have overfit to our data. So we add an instrument that uses no correctness labels. We fit the Jacobian lens [4] on generic text. It reads an activation for what it pushes the model to say next. We score each state by how far it leans toward “wrong” over “correct.” On dense models the lens matches the probe. On Mistral it reads 0.82 on MATH and 0.75 on MMLU-Pro, against the probe’s 0.83 and 0.75, and it reads 0.5 at the input layer, as expected. Two instruments built from different evidence find the same direction. And because the lens reads through the model’s own output pathway, that direction sits where it could influence what the model says.

Where the lens fails. The lens is a fixed dictionary from hidden-state directions to words, built once from generic text. On a dense model one dictionary is enough, because every input takes the same path to the output. GLM routes each input through a different subset of experts, and Qwen3.6 mixes two kinds of attention layers, so there is no single path and the lens averages many into a blur. The numbers show it: on GLM the lens’s entries for “correct” and “wrong” point almost the same way, cosine 0.96, against 0.40 on a dense model, and it scores 0.31 there and 0.43 on Qwen3.6, while the probe still works on both. Swap Qwen3.6 for Qwen2.5-14B, same family but dense, and the lens jumps to 0.88. The signal is there in every model. A label-free reader works only where the architecture gives it one path.

6 What this means

Practical use. Without an exact checker, the probe is the best per-attempt readout we found. It works where the model’s confidence is useless.

Limitations. A probe could be reading properties of failed output rather than a self-assessment. The surface baseline bounds this without closing it, most loosely on MATH, geometry (Appendix C), and GLM. Every number is a single seed. The causal result holds on Mistral. On GLM erasing does nothing, and on Qwen3.6 the self-report is too uninformative for erasing to test, though amplifying moves it. Gemma-4 leaves too few failures to test. Whether preference tuning creates the gap is open. A base model would answer it.

Takeaway. Four open models carry information about their own errors that their self-report does not. One uses it when it self-reports. In two, erasing it changes nothing.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarized first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$? The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True), the probability the model assigns to “True” when asked whether its answer is correct, 0.32; mean answer log-probability -0.05 .*

213 A single arithmetic slip inside a fluent derivation. Having just made the error, the model
 214 revises its stated confidence upward, while $P(\text{True})$ on the same transcript assigns only
 215 0.32 probability to correctness. The rest of this appendix gives the prompts and the full
 216 transcript.

217 A.1 Prompts

218 Each record is four forward passes: the three turns of Section 2 plus one read taken beforehand,
 219 at the last prompt token of the bare question, before any elicitation. The confidence turns
 220 run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with
 221 thinking on for hybrid reasoners. The fixed **Confidence: N** format is what makes the read
 222 site content-neutral: the digit token has identical local context on every record, so layer-0
 223 activations carry no signal by construction.

224 System prompt (MATH).

225 Solve the problem step by step. Put your final answer in `\boxed{\}`.

226 **Turn 1, prospective confidence.** The problem is followed by:

227 Before answering, estimate how likely you are to answer this question
 228 correctly. Reply with EXACTLY one line and nothing else:
 229 Confidence: N
 230 where N is an integer from 0 (certain to be wrong) to 100 (certain to be
 231 correct). Do NOT answer yet.

232 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

233 Now give your final answer inside `\boxed{\}`.

234 **Turn 3, retrospective confidence.** The attempt is left in context, then:

235 Now assess the answer you just gave. On a scale of 0 to 100, how confident
 236 are you that it is correct? Reply with exactly one line and nothing else:
 237 Confidence: N

238 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token
 239 mass on True versus False:

240 Consider your final answer above. Is it correct? Reply with exactly one word:
 241 True or False.

242 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
 243 assessment.

244 A.2 One record, verbatim

245 Record `math_test_number_theory_753`, sample 2 of 5.

246 **Problem.** What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?

247 **Turn 1 reply.** Confidence: 90

248 **Turn 2 reply (complete).**

249 First, we need to find the remainder when -11213141 is divided by 18.
 250 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

251 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

252 where r is the remainder.

253 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

254

$$r = -11213141 + 11213154$$

255

$$r = 3$$

256 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

257 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

258 **Turn 3 reply. Confidence: 95**

259 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
 260 11213154 checks out, and the model has written the correct subtraction on the line above its
 261 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing in
 262 the presentation marks the slip. The derivation is well formed, correctly structured, and
 263 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
 264 attempt that failed.

265 **Readouts of this attempt.**

readout	value
stated confidence, prospective (turn 1)	90
stated confidence, retrospective (turn 3)	95
P(True)	0.32
mean answer log-probability	−0.05
graded outcome	incorrect

267 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
 268 representative cell-level numbers are in Table 3, Appendix D.

269 B Calibration beyond ECE, and cell-level multiplicity

270 ECE depends on binning and on sample composition, so we report two binning-free proper
 271 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
 272 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
 273 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,
 274 worse calibrated in absolute terms. The steering results in Table 2 are reported as ECE
 275 because the steered runs stored per-gain aggregates rather than per-record confidences;
 276 recomputing Brier and NLL at each gain requires re-running the intervention and is the first
 277 thing we would add.

278 The cell-level bootstrap counts in Table 3 are uncorrected. With 16 cells, some individually
 279 significant results are expected under the null, so the cell counts should be read as descriptive.
 280 The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells
 281 with one tie, significantly positive in 10, and never significantly negative. A hierarchical
 282 model pooling across cells with model and domain as random effects is the right test and is
 283 not yet run.

284 C Alternative explanations for the probe signal

285 A probe could key on properties of failed derivations rather than on a dedicated metacognitive
 286 variable: low token probability, hedging language, or, on MATH, an early-layer channel
 287 next to the answer text. Our surface-features baseline (answer length, digit count, line
 288 count, word count, brace count) bounds this without closing it, and the bound is domain
 289 dependent: adding the probe to the surface features raises AUROC by +0.21 on MMLU-Pro
 290 but only +0.07 on MATH and −0.11 on geometry, where a construction that fails to compile

is visibly malformed. Two results argue against a pure surface account on the domains where the signal is strong. A direction fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs MATH’s self-report under ablation (0.880 $\rightarrow$ 0.709). And the pre-attempt read, taken where the activation is byte-identical across a question’s attempts, cannot recover the post-attempt direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would predict it could. A linear probe at one token also undercounts nonlinear or multi-token signal, so the reported numbers are a floor.

D Per-cell numbers and specificity controls

Table 3: Post-attempt correctness discrimination (AUROC). Left: self-report vs. the probe in representative cells, and the same comparison holding the question fixed (“within”), where the model’s yardstick is $P(\text{True})$ across the five attempts at one question; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post $\rightarrow$ pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model $\times$ domain	self-report	probe	within: $P(\text{True})$	within: probe
Gemma-4 $\times$ MATH	0.57	0.78	–	–
Qwen3.6 $\times$ MMLU-Pro	0.67	0.84	0.47	0.74
GLM $\times$ MMLU-Pro	0.67	0.85	0.59	0.73
Mistral $\times$ MATH	0.79	0.83	0.59	0.72

Mistral	pre	post	resid.	post $\rightarrow$ pre
MMLU-Pro	0.568	0.754	0.751	0.490
MATH	0.792	0.834	0.701	0.624
GPQA	0.593	0.572	0.544	0.479
GeoGen	0.770	0.676	0.505	0.648

E Checkpoints

All models are public Hugging Face releases, run in bfloat16 without quantization, greedy decoding: `mistralai/Mistral-Small-24B-Instruct-2501`, `Qwen/Qwen3.6-27B`, `zai-org/GLM-4.7-Flash`, `google/gemma-4-26B-A4B-it`; the dense within-family control for the lens is `Qwen/Qwen2.5-14B-Instruct`. Probe read layers, 70% of depth: Mistral 28 of 40, GLM 33 of 47, Gemma-4 21 of 30, Qwen3.6 45 of 64.

Why 70%. A pilot probed every layer and found the correctness signal rises through the network, plateaus from about half to 90% of depth, and drops in the last few layers, where the state is being turned into output tokens. Picking the best layer per cell inflated AUROC by 0.05 to 0.13, so we fixed one depth in advance for every model and every cell. Seventy percent is the middle of that plateau. The choice matches where prior work finds truth probes strongest [1, 7, 9], and the ablation checked it after the fact: on Mistral the same intervention at 30% of depth does nothing (Section 4).